Using deep learning for ultra-fast identification of V(D)J segments

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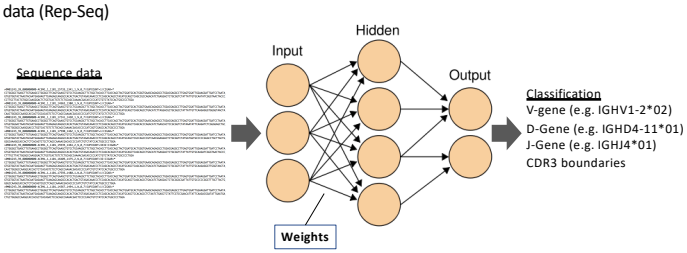
Abstract

Antibody repertoire (Rep-Seq) profiling of B-cells typically requires computational analysis of antibody sequences that have undergone V(D)J recombination and somatic hypermutation. An essential step in Rep-Seq bioinformatics is to determine the original V, D and J segments and their boundaries within the sequence. Widely used techniques such as IMGT and IgBLAST are computationally intense, and newly proposed methods based on probability models even more so, usually taking many hours to complete. Deep learning methods offer the possibility of separating the model training (computationally intense) from the classification step, which is very fast. Furthermore, the training process can be performed beforehand thus enabling the V(D)J classification to be performed rapidly by the end user as part of a Rep-Seq bioinformatics pipeline. Deep learning requires very large amounts of high quality data, but these are now becoming available as more Rep-Seq datasets are published. We have developed a deep learning approach to enable ultra-fast processing of Rep-Seq data.

Introduction

Deep learning (artificial neural network) techniques can be used to classify IGHV sequence data (Rep-Seq)



1. Training phase involves adapting the weights so that the model will classify each sequence correctly. This phase is relatively intense computationally.
2. Once trained correctly, the neural network can perform classifications efficiently.

Network diagram from http://en.wikipedia.org/wiki/Artificial_neural_network

Results

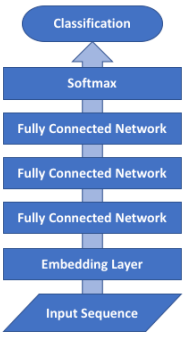
Input data

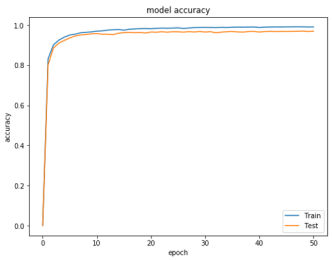
288,773 high quality (UMI barcoded) human IGHV sequences, each derived from a distinct clone (unique CDR3). Sequences were classified beforehand using IMGT/High-Vquest. 80% of sequences used for training, 20% for evaluation (test set).

Task: Classify each sequence into one of the 63 human IGHV genes in IMGT (not alleles yet).

Network configurations

1. Feed-forward network

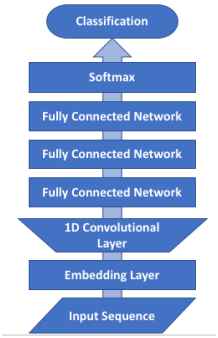


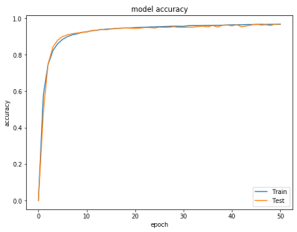


	Final accuracy
Training set	99.14%
Test set	96.96%
Epochs to reach 95% accuracy: 8	

Speed: Trained network can process 100,000 sequences in 9.02 seconds

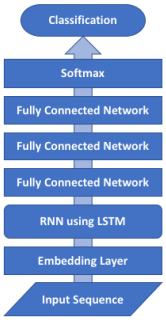
2. Convolutional network

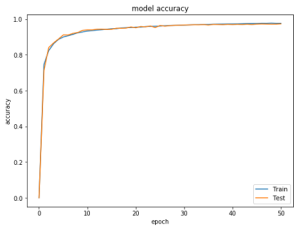




	Final accuracy
Training set	96.75%
Test set	96.73%
Epochs to reach 95% accuracy: 18	

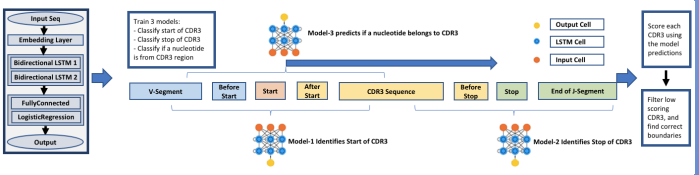
3. Recurrent neural network





	Final accuracy
Training set	97.73%
Test set	97.42%
Epochs to reach 95% accuracy: 15	

Identification of CDR3 boundaries



Model	Validation Dataset		Precision	Recall	F1-Score
Start of CDR3	Original	100000	98%	100%	99%
	Negative	10499970	100%	100%	100%
Stop of CDR3	Original	100000	75%	100%	86%
	Negative	10499970	100%	100%	100%
Inside CDR3	Original	3942762	93%	98%	96%
	Negative	3942762	98%	93%	95%

Conclusions

1. Neural networks (deep learning) can be trained and used for efficient V-gene classification.
2. Test set accuracy was high for all network configurations, but highest for recurrent network.
3. No evidence for overfitting with convolution or recurrent networks.
4. Ultra-fast classification: trained network can process >10,000 sequences per second.

Program for fast V-gene classification: http://www.ams.sunysb.edu/~maccarth/software.html

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